

Salamander Lemma on polycomplexes

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1 Introduction

Adottiamo un sistema di indici interi per descrivere un n -cocomplesso. Data una categoria abeliana $\mathcal{A}$ consideriamo una famiglia di oggetti indicizzata su $\mathbb{Z}^n$

$$A = \{A_x\}_{x \in \mathbb{Z}^n}, \quad A_x \in \text{Ob}(\mathcal{A})$$

Siano $e_i \in \mathbb{Z}^n$ l' i -esimo vettore della base canonica di $\mathbb{Z}^n$. Consideriamo n famiglie di differenziali fra gli oggetti

$$\partial^{(i)} = \{\partial_x^{(i)}\}_{x \in \mathbb{Z}^n},$$

per ogni direzione $i \in \{1, \dots, n\}$ dove

$$\partial_x^{(i)} \in \text{Hom}_{\mathcal{A}}(A_x, A_{x+e_i})$$

tali che:

1. ogni riga è un complesso di cocatene:

$$\partial_{x+e_i}^{(i)} \circ \partial_x^{(i)} = 0, \quad \forall x \in \mathbb{Z}^n, \forall i \in \{1, \dots, n\}$$

2. commutatività del diagramma:

$$\partial_{x+e_i}^{(j)} \circ \partial_x^{(i)} = \partial_{x+e_j}^{(i)} \circ \partial_x^{(j)}, \quad \forall x \in \mathbb{Z}^n, i \neq j$$

Scriveremo $\pm$ per denotare il numero ± 1 . Una coordinata $(\alpha_1, \dots, \alpha_n)$ può essere scritta come $\alpha_1, \dots, \alpha_n$. Una coordinata $(\alpha, \dots, \alpha)$ verrà denotata semplicemente con α . Denotiamo pure sempre $A = A_0$.

Definition 1.1. Se $x \leq y$ coordinata per coordinata, definiamo $\partial[x, y]: A_x \rightarrow A_y$ come la composizione lungo un qualsiasi cammino monotono da x a y . Per commutatività dei quadrati elementari, tale composizione è indipendente dal cammino scelto.

Definition 1.2. Per ogni $x \in \mathbb{Z}^n$, e per ogni due famiglie finite $L \subseteq \mathbb{Z}_{\leq 0}^n \setminus \{0\}^n$, $U \subseteq \mathbb{N}^n \setminus \{0\}^n$ tali che abbia senso, poniamo

$$I_L(x) \triangleq \sum_{\lambda \in L} \text{im } \partial[x + \lambda, x], \quad K_U(x) \triangleq \bigcap_{\mu \in U} \ker \partial[x, x + \mu]$$

Quando vale

$$I_L(x) \subseteq K_U(x)$$

definiamo

$$H_L^U(x) \triangleq \frac{K_U(x)}{I_L(x)}$$

Concretamente, per gli oggetti comologici centrati su un oggetto centrale A_0 con struttura $L = \{l_i\}_{i=1}^h \subseteq \{-1, 0\}^n$ e $U = \{u_i\}_{i=1}^k \subseteq \{0, +1\}^n$ abbiamo

$$H_L^U = \frac{\bigcap_{i=1}^k \ker \partial[0, U_i]}{\sum_{i=1}^h \text{im } \partial[L_i, 0]}$$

quando esiste. Ci stiamo quindi restringendo ad un intorno locale dell'oggetto centrale.

Nel paper di Bergman viene notato che il reticolo di oggetti comologici viene generato dai down-set. In generale questo numero combinatorico è il numero di Dedekind $M(n)$. Tuttavia, vengono scartati i casi banali del down-set vuoto e del down-set totale. Quindi in generale abbiamo

- $n = 1$: $M(1) - 2 = 1$
- $n = 2$: $M(2) - 2 = 4$
- $n = 3$: $M(3) - 2 = 18$

- $n = 4$: $M(4) - 2 = 166$
- $n = 5$: $M(5) - 2 = 7579$

Cominciamo generalizzando il concetto di *donor* e *receptor* come i gruppi comologici perfettamente diagonali che attraversano l'oggetto centrale.

Definition 1.3. *The n -receptor centered at x is defined as*

$$R(x) \triangleq \frac{\bigcap_{i=1}^n \ker \partial[x, x + e_i]}{\operatorname{im} \partial[x - 1, x]} = H_{-1}^{e_1, \dots, e_n}(x)$$

Definition 1.4. *The n -donor centered at x is defined as*

$$D(x) \triangleq \frac{\ker \partial[x, x + 1]}{\sum_{i=1}^n \operatorname{im} \partial[x - e_i, x]} = H_{-e_1, \dots, -e_n}^1$$

Definition 1.5. *The i -axial cohmology centered at x is defined as*

$$H_i(x) \triangleq \frac{\ker \partial[x, x + e_i]}{\operatorname{im} \partial[x - e_i, x]} = H_{-e_i}^{e_i}$$

for $i \in \{1, \dots, n\}$.

Cominciamo dimostrando esplicitamente il lemma 1.4 del paper originale per poi generalizzarlo.

Lemma 1.1. *Each arrow $f: A \rightarrow B$ in a double complex induces an arrow $A_\square \rightarrow {}^\square B$:*

Proof. Without loss of generality, let us consider the horizontal case:

$$\begin{array}{ccccccc} & & \bullet & \xrightarrow{\quad} & \bullet & & \\ & & \downarrow b & \searrow p & \downarrow & & \\ \bullet & \xrightarrow{a} & A & \xrightarrow{f} & B & \xrightarrow{d} & \bullet \\ & & \downarrow & \searrow q & \downarrow c & & \\ & & \bullet & \xrightarrow{\quad} & \bullet & & \end{array}$$

$$A_\square = \frac{\ker q}{\operatorname{im} a + \operatorname{im} b}, \quad {}^\square B = \frac{\ker c \cap \ker d}{\operatorname{im} p}$$

Let

$$K := \ker q, \quad M := \operatorname{im} a + \operatorname{im} b, \quad H := \ker c \cap \ker d, \quad N := \operatorname{im} p$$

we need to show that $f: A \rightarrow B$ induces a map $K/M \rightarrow H/N$. We need to ensure that the quotients are well-defined, meaning that $\operatorname{im} a + \operatorname{im} b \subseteq \ker q$ and $\operatorname{im} p \subseteq \ker c \cap \ker d$. Since the rows and columns are complexes and the squares commute, these inclusions hold.

- *f sends $\ker q$ into $\ker c \cap \ker d$:* let $\iota_K: \ker q \hookrightarrow A$ and $\iota_H: \ker c \cap \ker d \hookrightarrow B$ be the canonical inclusions. By definition of q and $\ker q$, $q\iota_K = cf\iota_K = 0$. By the properties of the complex, $df = 0$ and thus $df\iota_K = 0$. Thus, $f\iota_K: K \rightarrow B$ is nullified both by d and c . By the universal property of the intersection of kernels, meaning the pullback of $\ker c \cap \ker d$, there exists a unique morphism $\tilde{f}: K \rightarrow H$ such that $\iota_H \tilde{f} = f\iota_K$. Thus, f is restricted to $\tilde{f}: \ker q \rightarrow \ker c \cap \ker d$.
- *f sends $\operatorname{im} a + \operatorname{im} b$ into $\operatorname{im} p$:* we can separately check $\operatorname{im} a$ and $\operatorname{im} b$. By the properties of the complex, $fa = 0$ meaning that fa factors through the subobject $\operatorname{im} p \hookrightarrow B$ via the zero morphism. By definition of p , $fb = p$ and thus the image of fb is contained in $\operatorname{im} p$. By the universal property of the sum of subobjects, $\operatorname{im} a + \operatorname{im} b$ is the smallest object of A containing both $\operatorname{im} a$ and $\operatorname{im} b$.

$$\begin{array}{ccccc} M & \hookrightarrow & K & \xrightarrow{\pi_K} & K/M \\ \downarrow & & \downarrow \tilde{f} & & \downarrow \bar{f} \\ N & \hookrightarrow & H & \xrightarrow{\pi_H} & H/N \end{array} \quad \text{with } \pi_K: K \rightarrow K/M \text{ and } \pi_H: H \rightarrow H/N$$

H/N canonical projections.

Since $\pi_H \tilde{f}$ vanishes on M , by the universal property of the cokernel of $M \hookrightarrow K$, there exists a unique morphism $\bar{f}: K/M \rightarrow H/N$ such that $\bar{f}\pi_K = \pi_H \tilde{f}$. Thus, there exists a unique map

$$\bar{f}: \frac{\ker q}{\text{im } a + \text{im } b} \rightarrow \frac{\ker c \cap \ker d}{\text{im } p}$$

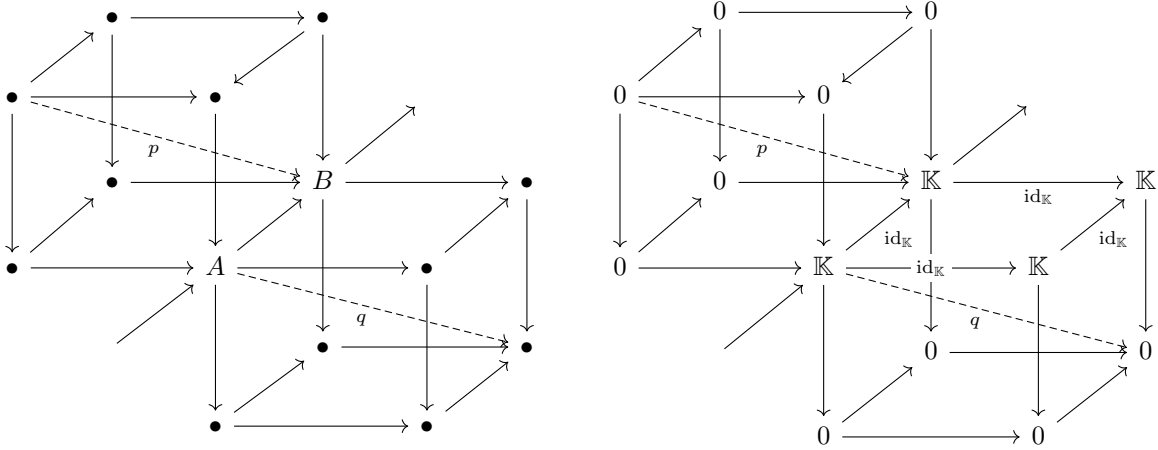
induced by f . □

Written using our notation, this lemma states that for $n = 2$, a morphism $\partial[x, x + e_i]$ induces a morphism $D(x) \rightarrow R(x + e_i)$ for every $i \in \{1, 2\}$. The same lemma does not generalize for $n > 2$.

Proof. Consider the following counterexample: we consider $\mathbf{Vect}_{\mathbb{K}}$ for some field $\mathbb{K}$. We construct a 3-cocomplex where all the objects are zero except for

$$A = \mathbb{K}, \quad A_{1,0,0} = \mathbb{K}, \quad A_{1,0,1} = \mathbb{K}, \quad A_{0,0,1} = \mathbb{K} = B$$

All the maps are the null maps, except the maps between the latter objects which are the identity $\text{id}_{\mathbb{K}}$.



The diagram commutes as the compositions of the non-trivial square are still the identity. We will now compute the donor of A , $D(0)$ and the receiver of B , $D(e_3)$.

1. *donor of A*: we have

$$D(0) = \frac{\ker \partial[0, 1]}{\text{im } \partial[-e_1, 0] + \text{im } \partial[-e_2, 0] + \text{im } \partial[-e_3, 0]}$$

In particular, every monotone path from coordinates 0 to 1 must go through the second dimension (downwards), which is the zero morphism. Thus, $\partial[0, 1] = 0$. This means that every element of $\mathbb{K}$ is sent to 0 and thus $\ker \partial[0, 1] = \mathbb{K}$. Furthermore, $A_{-e_1} = A_{-e_2} = A_{-e_3} = 0$ meaning that

$$\text{im } \partial[-e_1, 0] + \text{im } \partial[-e_2, 0] + \text{im } \partial[-e_3, 0] = 0$$

Thus, $D(0) = \mathbb{K}$.

2. *receiver of B*: we have

$$R(e_2) = \frac{\ker \partial[e_3, e_3 + e_1] \cap \ker \partial[e_3, e_3 + e_2] \cap \ker \partial[e_3, 2e_3]}{\text{im } \partial[e_3 - 1, e_2]}$$

For the kernels we have

$$\ker \partial[e_3, e_3 + e_1] \cap \ker \partial[e_3, e_3 + e_2] \cap \ker \partial[e_3, 2e_3] = \ker 0 \cap \ker 0 \cap \ker \text{id}_{\mathbb{K}} = 0$$

Furtermore, $A_{e_3-1} = A_{-1,-1,0} = 0$, and thus $\text{im } p = 0$. Thus, $R(e_3) = 0$.

Suppose that $f = \text{id}_{\mathbb{K}}: A \rightarrow B$ induces a morphism $D(0) \rightarrow R(e_3)$. Then, f must send elements of the numerator of $D(0)$ into the numerator of $R(e_3)$, meaning

$$f(\ker \partial[0, 1]) \subseteq \ker \partial[e_3, e_3 + e_1] \cap \ker \partial[e_3, e_3 + e_2] \cap \ker \partial[e_3, 2e_3]$$

However, $\ker \partial[0, 1] = \mathbb{K}$ and $f = \text{id}_{\mathbb{K}}: A_0 \rightarrow A_{e_3}$ meaning that f does not factor through the zero subobject $0 \hookrightarrow A_{e_3}$. In particular the arrow $f: A_0 \rightarrow A_{e_3}$ does not restrict to an arrow

$$\ker \partial[0, 1] \rightarrow \ker \partial[e_3, e_3 + e_1] \cap \ker \partial[e_3, e_3 + e_2] \cap \ker \partial[e_3, 2e_3]$$

Thus, f does not induce such a map. □

1.1 Caso monodimensionale

Per i complessi di cocatene consideriamo

$$A(-) \xrightarrow{\partial[-,0]} A \xrightarrow{\partial[0,+]} A(+)$$

Possiamo solo definire l'oggetto coomologico classico

$$H_{-}^{+} \triangleq \frac{\ker \partial[0, +]}{\text{im } \partial[-, 0]}$$

1.2 Caso bidimensionale

For the cochain bicomplexes we consider the following local commutative diagram

$$\begin{array}{ccccc} A(-) & \xrightarrow{\partial[(-),(0,-)]} & A(0, -) & \xrightarrow{\partial[(0,-),+]} & A(+) \\ \downarrow \partial[(-),(-,0)] & & \downarrow \partial[(0,-),0] & & \downarrow \partial[(+),(+,0)] \\ A(-, 0) & \xrightarrow{\partial[(-,0),0]} & A & \xrightarrow{\partial[0,(+,0)]} & A(+, 0) \\ \downarrow \partial[(-,0),-] & & \downarrow \partial[0,(0,+)] & & \downarrow \partial[(+,0),+] \\ A(-) & \xrightarrow{\partial[-,(0,+)]} & A(0, +) & \xrightarrow{\partial[(0,+),+]} & A(+) \end{array}$$

We define the directional homological objects as follows

$$\begin{array}{ll} \begin{array}{c} \bullet \quad \square \quad \bullet \\ \square \end{array} & H_{-,0}^{+,0} \triangleq \frac{\ker \partial[0, (+, 0)]}{\text{im } \partial[(-, 0), 0]} \\ \begin{array}{c} \square \quad \bullet \\ \square \end{array} & H_{-,0}^{+,0} \triangleq \frac{\ker \partial[0, (0, +)]}{\text{im } \partial[(0, -), 0]} \\ \begin{array}{c} \bullet \quad \square \quad \bullet \\ \square \end{array} & H_{-}^{(0,+),(+,0)} \triangleq \frac{\ker \partial[0, (+, 0)] \cap \ker \partial[0, (0, +)]}{\text{im } \partial[-, 0]} \\ \begin{array}{c} \square \quad \bullet \\ \bullet \quad \square \end{array} & H_{(-,0),(0,-)}^{+,-} \triangleq \frac{\ker \partial[0, +]}{\text{im } \partial[(-, 0), 0] + \text{im } \partial[(0, -), 0]} \end{array}$$

Abbiamo il diagramma parziale

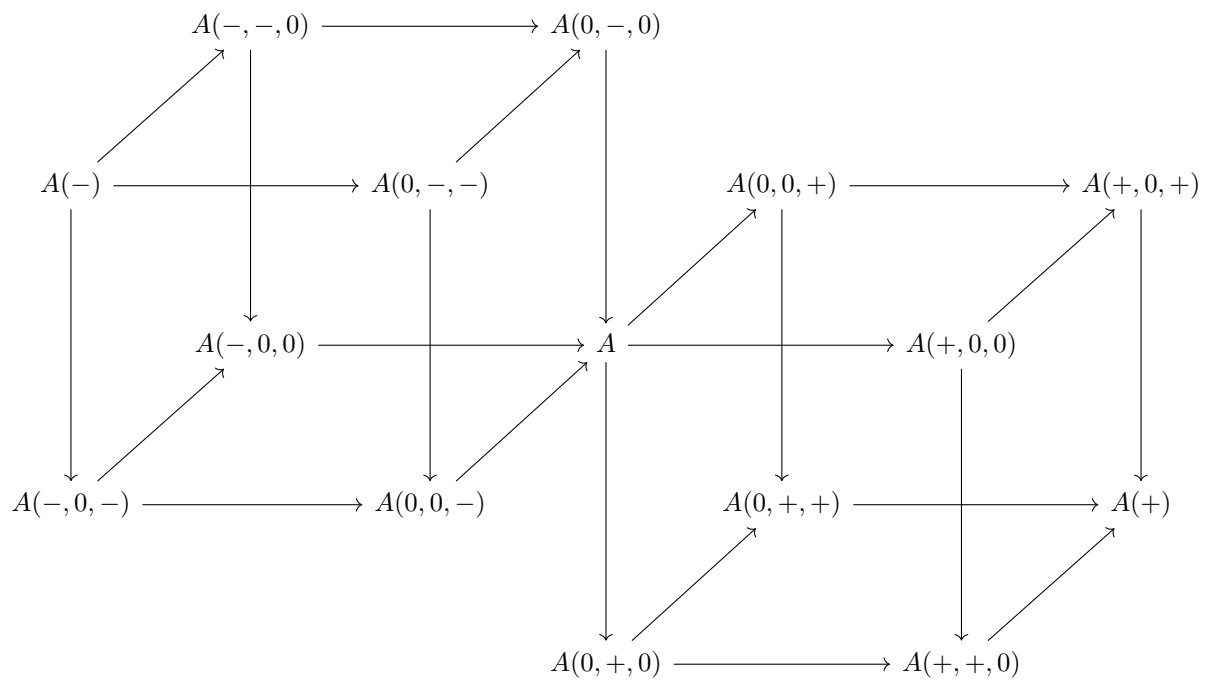


Figure 1 displays 18 diagrams illustrating the evolution of a 2D Ising spin configuration on a 2x2 grid of cubes. Each diagram is labeled (1) through (18). The cubes are arranged in a 2x2 grid, and the spin configuration (black dot for up, white for down) evolves from (1) to (18).

- (1) $H_{(-,0,0)}^{(+,0,0),(0,0,+)} \triangleq \frac{\ker \partial[0, (+, 0, 0)] \cap \ker \partial[0, (0, 0, +)]}{\operatorname{im} \partial[(-, 0, 0), 0]}$
- (2) $H_{(-,-,0)}^{(+,0,0),(0,+,0)} \triangleq \frac{\ker \partial[0, (+, 0, 0)] \cap \ker \partial[0, (0, +, 0)]}{\operatorname{im} \partial[(-, -, 0), 0]}$
- (3) $H_{(0,-,-)}^{(0,+,+),(0,0,+)} \triangleq \frac{\ker \partial[0, (0, +, 0)] \cap \ker \partial[0, (0, 0, +)]}{\operatorname{im} \partial[(0, -, -), 0]}$
- (4) $H_{(-,-,0),(0,-,0)}^{(+,0,0),(0,+,+)} \triangleq \frac{\ker \partial[0, (+, 0, 0)] \cap \ker \partial[0, (0, +, +)]}{\operatorname{im} \partial[(-, -, 0), 0] + \operatorname{im} \partial[(-, 0, -), 0]}$
- (5) $H_{(-,-,0),(0,-,-)}^{(+,0,0),(0,+,+)} \triangleq \frac{\ker \partial[0, (+, 0, 0)] \cap \ker \partial[0, (0, +, +)]}{\operatorname{im} \partial[(-, -, 0), 0] + \operatorname{im} \partial[(0, -, -), 0]}$
- (6) $H_{(-,0,-),(0,-,-)}^{(+,0,0),(0,+,+)} \triangleq \frac{\ker \partial[0, (+, 0, 0)] \cap \ker \partial[0, (0, +, +)]}{\operatorname{im} \partial[(-, 0, -), 0] + \operatorname{im} \partial[(0, -, -), 0]}$
- (7) $H_{(0,-,-),(0,-,0)}^{(+,+,0),(+,0,+)} \triangleq \frac{\ker \partial[0, (+, +, 0)] \cap \ker \partial[0, (+, 0, +)]}{\operatorname{im} \partial[(0, -, -), 0] + \operatorname{im} \partial[(-, 0, 0), 0]}$
- (8) $H_{(0,-,-),(0,-,0)}^{(+,+,0),(0,+,+)} \triangleq \frac{\ker \partial[0, (+, +, 0)] \cap \ker \partial[0, (0, +, +)]}{\operatorname{im} \partial[(0, -, -), 0] + \operatorname{im} \partial[(-, 0, 0), 0]}$
- (9) $H_{(0,-,-),(0,-,0)}^{(0,+,+),(+,0,+)} \triangleq \frac{\ker \partial[0, (0, +, +)] \cap \ker \partial[0, (+, 0, +)]}{\operatorname{im} \partial[(0, -, -), 0] + \operatorname{im} \partial[(-, 0, 0), 0]}$
- (10) $H_{(0,0,-),(0,-,0)}^{(+,0,+)} \triangleq \frac{\ker \partial[0, (+, 0, +)]}{\operatorname{im} \partial[(0, 0, -), 0] + \operatorname{im} \partial[(-, 0, 0), 0]}$
- (11) $H_{(0,0,-),(0,-,0)}^{(0,+,+)} \triangleq \frac{\ker \partial[0, (0, +, +)]}{\operatorname{im} \partial[(0, 0, -), 0] + \operatorname{im} \partial[(0, -, 0), 0]}$
- (12) $H_{(-,0,0),(0,-,0)}^{(+,+,0)} \triangleq \frac{\ker \partial[0, (+, +, 0)]}{\operatorname{im} \partial[(-, 0, 0), 0] + \operatorname{im} \partial[(0, -, 0), 0]}$
- (13) $H_{(0,0,-),(0,-,0),(0,-,0)}^{+} \triangleq \frac{\ker \partial[0, +]}{\operatorname{im} \partial[(0, 0, -), 0] + \operatorname{im} \partial[(-, 0, 0), 0] + \operatorname{im} \partial[(0, -, 0), 0]}$
- (14) $H_{(-,0,-),(0,-,-),(0,-,0)}^{(+,+,0),(0,+,+),(+,0,+)} \triangleq \frac{\ker \partial[0, (+, +, 0)] \cap \ker \partial[0, (0, +, +)] \cap \ker \partial[0, (+, 0, +)]}{\operatorname{im} \partial[(-, 0, -), 0] + \operatorname{im} \partial[(0, -, -), 0] + \operatorname{im} \partial[(-, -, 0), 0]}$
- (15) $H_{-}^{(0,+,0),(+,0,0),(0,0,+)} \triangleq \frac{\ker \partial[0, (0, +, 0)] \cap \ker \partial[0, (+, 0, 0)] \cap \ker \partial[0, (0, 0, +)]}{\operatorname{im} \partial[-, 0]}$
- (16) $H_{-,0,0}^{+,0,0} \triangleq \frac{\ker \partial[0, (+, 0, 0)]}{\operatorname{im} \partial[(-, 0, 0), 0]}$
- (17) $H_{0,-,0}^{0,+,0} \triangleq \frac{\ker \partial[0, (0, +, 0)]}{\operatorname{im} \partial[(0, -, 0), 0]}$
- (18) $H_{0,0,-}^{0,0,+} \triangleq \frac{\ker \partial[0, (0, 0, +)]}{\operatorname{im} \partial[(0, 0, -), 0]}$

2 Altro

Caso verticale del diagramma del bicomplesso.

$$\begin{array}{ccccc}
 & & U & & \\
 & & \downarrow c & & \\
 L & \xrightarrow{d} & A & \xrightarrow{e} & C \\
 \downarrow b & & \downarrow v & & \downarrow g \\
 M & \xrightarrow{k} & B & \xrightarrow{h} & D \\
 & & \downarrow m & & \\
 & & N & &
 \end{array}$$

$$D(A) = \frac{\ker(h \circ v)}{\text{im } c + \text{im } d} = \frac{\ker(g \circ e)}{\text{im } c + \text{im } d}, \quad R(B) = \frac{\ker h \cap \ker m}{\text{im}(v \circ d)} = \frac{\ker h \cap \ker m}{\text{im}(k \circ b)}.$$

2.1 Casi tridimensionali

Ci piacerebbe dire che un morfismo $A \rightarrow B$ induce un morfismo $D(A) \rightarrow D(B)$ questo già non è vero per $n=3$. Controesempio sugli spazi vettoriali. Che cosa ci servirebbe? Ci servirebbe un donor che ha l'intersezione dei 3 kernel a due step. Ma questo è l'elemento 14, infatti dati $A \rightarrow B \rightarrow C$ abbiamo due mappe indotte $D(A) \rightarrow M(B) \rightarrow R(C)$. Abbiamo quindi $3 * 2 = 6$ casi, ma per la commutatività metà coincidono quindi 3.

E questo invece dovrebbe essere le intramural maps.

